Azure Sands Resort & Spa

Last Resort

GROUP 10

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Project Overview

Problem & Scope

What the System Manages?

Reservations & Guest Stays

Room & Facility Resources

Events & Bookings

Billing & Payments

Entity Relationship Diagram

OVERVIEW OF ERD DOMAINS

The ERD consists of five domains: Physical Hierarchy, People, Reservations and Stays, Events, and Billing, organized into 23 tables.

(hotel_complex, building, wing, floor, room, room_bed, room_adjacency, bed_type · 8 tables), People (person, organization, guest, billed_party · 4 tables), Reservations & Stays (reservation, reservation_guest, reservation_room_request, stay, stay_room · 5 tables), Events (event, facility_booking · 2 tables), Billing (service_item, charge, payment, deposit · 4 tables)

All tables normalized to 3NF.each attribute depends solely on its primary key, no transitive dependencies.

Data Coverage Overview

PHYSICAL

The physical domain includes 1 hotel complex, comprising 2 buildings, which house 3 wings and 4 floors with a total of 6 rooms.

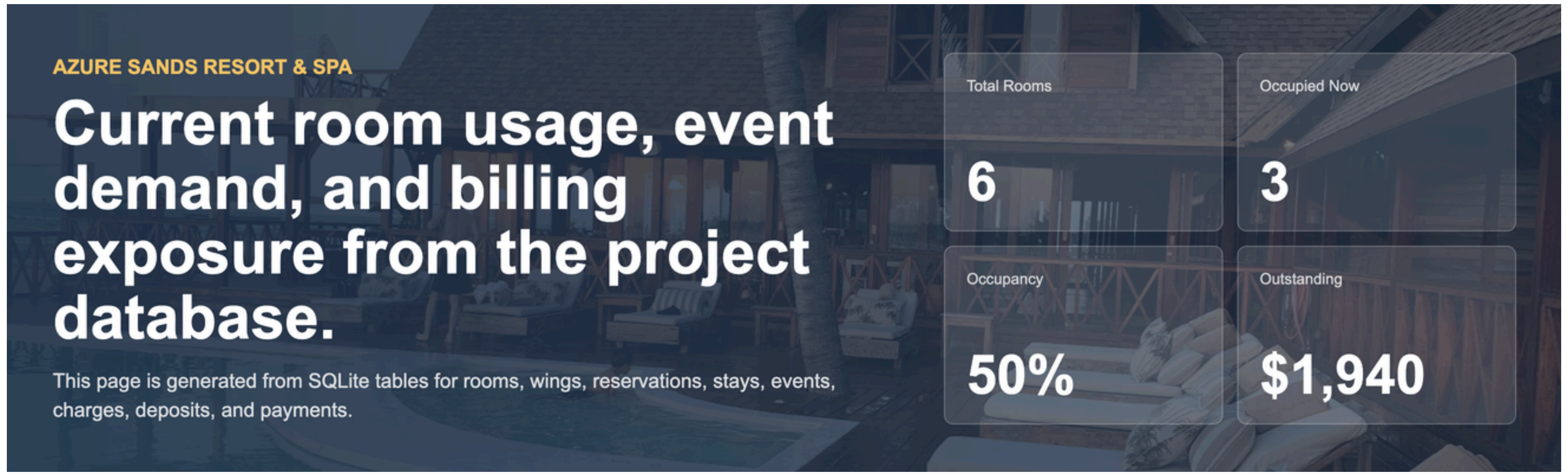
PEOPLE

In the people domain, there are 6 persons, 3 organizations, and 6 guests, along with 4 billed parties for effective management.

RESERVATIONS

This domain encompasses 4 reservations, 5 stays, and 5 stay room assignments, ensuring a comprehensive overview of guest accommodations.

Web Demo

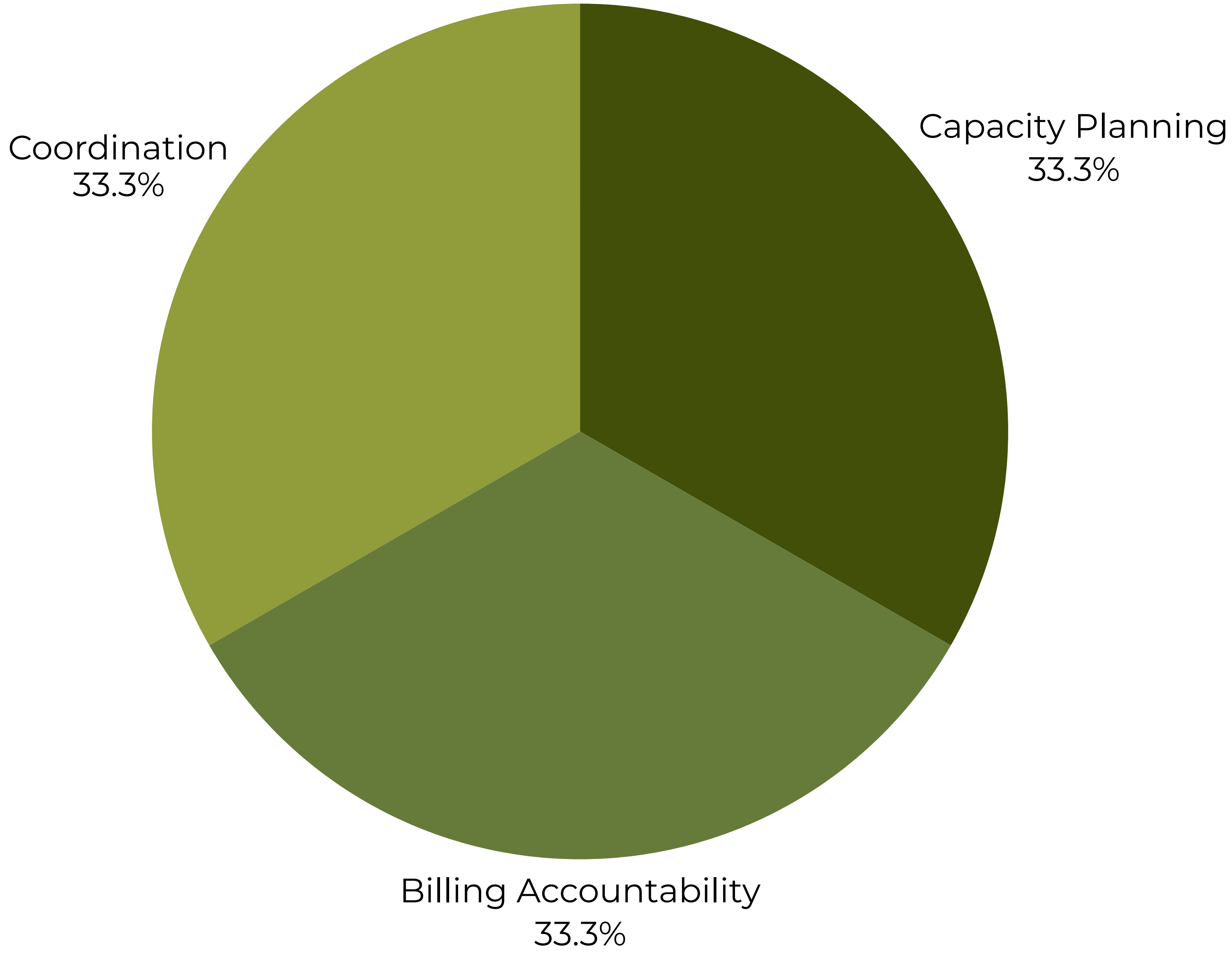


Occupied Room Count

Outstanding Balance by Billed Party

ID	BILLED PARTY	CHARGES	PAYMENTS	OUTSTANDING
2	NeuroLink Analytics	$2300	$1200	$1100
3	BlueWave Ventures	$1500	$800	$700
1	Elena Marquez	$240	$100	$140
4	Solstice Creative Group	$0	$0	$0

Charges, Payments, and Outstanding Balance



Our web demo is divided into three major sections, which, combined, covers every SQL query output

Tradeoffs & Limitations

Simplified for Project Scope

Dynamic Pricing
Base rates only
Pricing logic handled in application layer

Staff Scheduling & Access Control
Omitted as future extensions

Service Workflow
Simplified charge relationships to reduce complexity

Data & Design Decisions

Sample Data
Supports analytical queries but not full operational scale

Business Rules
Some handled through controlled data input rather than database constraints

Design Philosophy
Focus on correctness of relationships over exhaustive rule enforcement

Conclusion

What We Built

A relational database modeling complex resort operations (reservations, stays, events, facilities, and billing) through 23 normalized tables in 3NF

Key Strengths

Clear separation of reservations vs. stays and charges vs. payments

Physical hierarchy supports resource allocation and facility analysis

Analytical queries drive operational and financial decision-making through dashboards

Project Outcome

Demonstrates how database design supports both day-to-day coordination and strategic insight in hospitality management

Thank you for listening!